

Greedy Goody

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

You are given an array A of N values representing different costs for some items. And you want to buy exactly X items from them. and you are given a number T , and you are required to do the following:

- If T value equals to 1, then you have to choose the cheapest X items.
- If T value equals to 2, then you have to choose the most expensive X items.

You are required to print the total cost of the X items wheather it is the cheapest or the most expensive.

Input

You are given two lines of input:

The first line contains three values ($1 \leq N \leq 1000$) ($1 \leq T \leq 2$) ($0 \leq X \leq N$) the number of elements in the array A , the type of the operation you will do, and the number of most expensive / cheapest items.

The second line will have N numbers representing the array values $A_i, \dots, A_N$ ($1 \leq A_i \leq 100$).

Output

Just print the total cost of the X items.

Examples

standard input	standard output
5 1 3 1 2 3 4 5	6
5 2 3 1 2 3 4 5	12